

# GANESH GUPTA

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## Experience

### Oracle

Jun 2025 – Present

*Software Engineer, AI Platform*

- Own technical design end to end for distributed microservices exposing REST APIs at high request volume, sustaining 99%+ uptime under production load at sub-200ms p99 latency on multi-tenant AWS infrastructure.
- Engineered fault-tolerant traffic and workload routing logic across distributed services using Kafka event streams and pattern recognition, shifting load away from unhealthy nodes and mitigating undesired request behavior automatically.
- Designed stateful recovery via Redis checkpointing; interrupted distributed workflows resume from the last committed step across fault-tolerant execution.
- Built observability tooling replacing polling with event-driven monitoring, cutting redundant API calls 90% and enabling real-time reliability tracking; participate in on-call rotation for the platform.
- Drove a deployment-gated evaluation pipeline enforcing automated quality and reliability checks, blocking 15% of failed releases before production, and mentor engineers through code reviews and design write-ups.

### QureAI (AI Startup)

Mar 2026 – Present

*ML Infrastructure Engineer*

- Built a model observability framework tracking drift, latency, and evaluation regressions in production, enabling governance-aligned rollback workflows as an early engineer at this AI startup.
- Improved RAG retrieval recall 61% to 89% on a 2K-query eval set by rebuilding the chunking pipeline with hybrid dense-sparse retrieval and re-ranking.

### Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

*Software Engineer*

- Architected and operated 25+ production C++ and SQL-backed microservices in distributed systems for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Optimized SQL schema design and query performance, cutting nightly data aggregation from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Standardized failure-handling contracts (retry, timeout, propagation semantics) across 25+ services, reducing production escalations 60% in one quarter.
- Applied OOP principles and design patterns to build testable, maintainable backend components, mentoring engineers through code reviews and design write-ups.

## Projects

**LangFetch** | [tinyurl.com/LangFetch](https://tinyurl.com/LangFetch)

- LangGraph agent compiling natural-language questions into dependency-ordered SQL subqueries with Pinecone vector retrieval and hybrid re-ranking; deployed as an MCP-compatible server for enterprise data queries.

**Distributed KV Store / VPC Control Plane Simulator** | [github.com/ganeshhgupta/distributed-kv-store](https://github.com/ganeshhgupta/distributed-kv-store)

- Raft-consensus key-value store in C++ with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing for fault-tolerant distributed coordination.

**Watch-your-LLM** | [github.com/ganeshhgupta/watch-your-llm](https://github.com/ganeshhgupta/watch-your-llm)

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL.

## Education

**University of Texas at Arlington**

Aug 2023 – May 2025

*MS, Computer Science, GPA 4.0/4.0*

**MCKV Institute of Engineering**

Jul 2016 – Jul 2020

*BTech, Computer Science*

## Skills

**Languages:** Python, Java, C++, TypeScript, SQL

**Backend & APIs:** RESTful APIs, Distributed Systems, Microservices, Event-Driven Architecture, Concurrency

**Databases:** PostgreSQL, MongoDB, DynamoDB, Schema Design & Query Optimization

**Cloud & DevOps:** AWS (Lambda, EC2, RDS, SQS), Kubernetes, Kafka, Docker, Git, CI/CD

## Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer